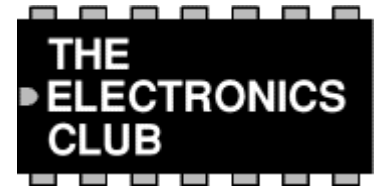


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Other Components

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Light Dependent Resistor (LDR)

An LDR is an **input transducer** (sensor) which converts brightness (light) to resistance. It is made from cadmium sulphide (CdS) and the resistance decreases as the brightness of light falling on the LDR increases.

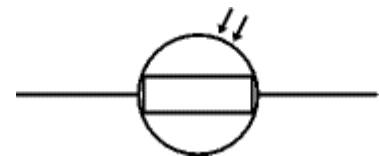
A [multimeter](#) can be used to find the resistance in darkness and bright light, these are the typical results for a standard LDR:

- **Darkness:** maximum resistance, about $1\text{M}\Omega$.
- **Very bright light:** minimum resistance, about 100Ω .

For many years the standard LDR has been the ORP12, now the NORPS12, which is about 13mm diameter. Miniature LDRs are also available and their diameter is about 5mm.



Photograph © [Rapid Electronics](#)



circuit symbol

An LDR may be connected either way round and no special precautions are required when soldering.

Thermistor

A thermistor is an **input transducer** (sensor) which converts temperature (heat) to resistance. Almost all thermistors have a negative temperature coefficient (NTC) which means their resistance decreases as their temperature increases. It is possible to make thermistors with a positive temperature coefficient (resistance increases as temperature increases) but these are rarely used. Always assume NTC if no information is given.

A [multimeter](#) can be used to find the resistance at various temperatures, these are some typical readings for example:

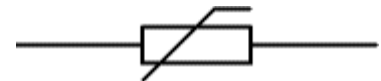
- **Icy water 0°C :** high resistance, about $12\text{k}\Omega$.
- **Room temperature 25°C :** medium resistance, about $5\text{k}\Omega$.
- **Boiling water 100°C :** low resistance, about 400Ω .



Photograph © [Rapid Electronics](#)

Suppliers usually specify thermistors by their resistance at 25°

C (room temperature). Thermistors take several seconds to respond to a sudden temperature change, small thermistors respond more rapidly.



circuit symbol

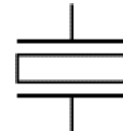
A thermistor may be connected either way round and no special precautions are required when soldering. If it is going to be immersed in water the thermistor and its connections should be insulated because water is a weak conductor; for example they could be coated with polyurethane varnish.

Piezo transducer

Piezo transducers are **output transducers** which convert an electrical signal to sound. They require a driver circuit (such as a 555 astable) to provide a signal and if this is near their natural (resonant) frequency of about 3kHz they will produce a particularly loud sound.



Photograph © [Rapid Electronics](#)



circuit symbol

Piezo transducers require a small current, usually less than 10mA, so they can be connected directly to the outputs of most ICs. They are ideal for buzzes and beeps, but are not suitable for speech or music because they distort the sound. They are sometimes supplied with red and black leads, but they may be connected either way round. PCB-mounting versions are also available.

Piezo transducers can also be used as **input transducers** for detecting sudden loud noises or impacts, effectively behaving as a crude microphone.

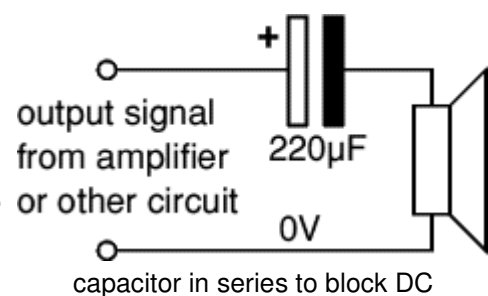
Loudspeaker

Loudspeakers are **output transducers** which convert an electrical signal to sound. Usually they are called 'speakers'. They require a driver circuit, such as a 555 astable or an audio amplifier, to provide a signal. There is a wide range available, but for many electronics projects a 300mW miniature loudspeaker is ideal. This type is about 70mm diameter and it is usually available with resistances of 8Ω and 64Ω . If a project specifies a 64Ω speaker you must use this higher resistance to prevent damage to the driving circuit.



Photograph © [Rapid Electronics](#)

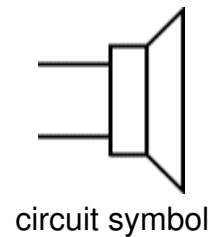
Most circuits used to drive loudspeakers produce an audio (AC) signal which is combined with a constant DC signal. The DC will make a large current flow through the speaker due to its low resistance, possibly damaging both the speaker and the driving circuit. To prevent this happening a large value electrolytic capacitor is connected in series with the speaker, this blocks DC but passes audio (AC) signals. See [capacitor coupling](#).



Loudspeakers may be connected either way round except in stereo circuits when the + and - markings on

their terminals must be observed to ensure the two speakers are in phase.

Correct polarity must always be observed for large speakers in cabinets because the cabinet may contain a small circuit (a 'crossover network') which diverts the high frequency signals to a small speaker (a 'tweeter') because the large main speaker is poor at reproducing them.



Miniature loudspeakers can also be used as a microphone and they work surprisingly well, certainly good enough for speech in an intercom system for example.

Buzzer and Bleeper

These devices are **output transducers** converting electrical energy to sound. They contain an internal oscillator to produce the sound which is set at about 400Hz for buzzers and about 3kHz for beepers.



Buzzer (about 400Hz)



Beeper (about 3kHz)

Photographs © [Rapid Electronics](#)

Buzzers have a voltage rating but it is only approximate, for example 6V and 12V buzzers can be used with a 9V supply. Their typical current is about 25mA.

circuit symbol



Beepers have wide voltage ranges, such as 3-30V, and they pass a low current of about 10mA.

Buzzers and beepers must be connected the right way round, their red lead is positive (+).

Inductor (coil)

An inductor is a coil of wire which may have a core of air, iron or ferrite (a brittle material made from iron). Its electrical property is called inductance and the unit for this is the henry, symbol H. 1H is very large so mH and μ H are used, $1000\mu\text{H} = 1\text{mH}$ and $1000\text{mH} = 1\text{H}$. Iron and ferrite cores increase the inductance. Inductors are mainly used in tuned circuits and to block high frequency AC signals (they are sometimes called chokes). They pass DC easily, but block AC signals, this is the opposite of [capacitors](#).



Inductor (miniature)



Ferrite rod

Photographs © [Rapid Electronics](#)

Inductors are rarely found in simple projects, but one exception is the tuning coil of a radio receiver. This is an inductor which you may have to make yourself by neatly winding enamelled copper wire around a ferrite rod.

Enamelled copper wire has very thin insulation, allowing the turns of the coil to be close together, but this makes it impossible to strip in the usual way - the best method is to gently pull the ends of the wire through folded emery paper.



circuit symbol

Warning: a ferrite rod is brittle so treat it like glass, not iron!

An inductor may be connected either way round and no special precautions are required when soldering.

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